Standard Operating Procedure (SOP)
Automated Identification of Diabetic Retinopathy from Fundus Images

1. Purpose
This SOP defines the standardized process for developing, training, and evaluating an artificial intelligence (AI) system to detect and classify Diabetic Retinopathy (DR) from retinal fundus images to enable early diagnosis and prevent vision loss.

2. Scope
This SOP applies to retinal image preparation, metadata handling, model development, training, evaluation, and validation for clinical AI research and medical device development.

3. Background
Diabetic Retinopathy is a microvascular complication of diabetes and a leading cause of preventable blindness. Early detection through automated screening systems can significantly reduce disease progression and healthcare burden.

4. Dataset Description
Total Images: 400
File Format: PNG
Max Image Size: 6 MB
Capture Method: Fundus Photography
Age Group: 35–65 years

5. DR Severity Labels
0 – No DR
1 – Mild
2 – Moderate
3 – Severe
4 – Proliferative DR

6. Metadata Requirements
Image ID, Gender, Age, Laterality, Image Quality, Camera Device ID

7. Data Privacy
All data is anonymized with no Personally Identifiable Information (PII). Dataset complies with medical data protection standards.

8. Data Preprocessing
Image resizing, normalization, quality filtering, metadata integration, and optional data augmentation are performed prior to training.

9. Model Development
Transfer learning using CNN architectures such as EfficientNet, ResNet, or MobileNet with five-class classification output.

10. Training & Evaluation
Weighted loss functions, Adam optimizer, learning rate scheduling, early stopping, and evaluation using accuracy, precision, recall, F1-score, and confusion matrix.

11. Expected Outcome
Early and accurate detection of Diabetic Retinopathy to support clinical decision-making and prevent avoidable vision loss.

12. Conclusion
This SOP ensures a standardized, privacy-compliant, and clinically relevant workflow for automated Diabetic Retinopathy screening.